AP Calculus BC Placement Diagnostic

Name: _____ Date: _____ Time used: _____ min

How this works. This is a no-calculator placement exam, 50 points total. Part I is 14 multiple-choice questions (1 point each). Part II is 4 free-response questions (9 points each). Budget about 75 minutes. Show every step in Part II, name any theorem you use, and write complete sentences when a question asks you to justify. It is built to find gaps, so skipping a question tells us as much as getting it wrong. Do not guess wildly; mark any question you are unsure about with a star.

Part I: Multiple Choice

1. Evaluate $\lim_{x \to 0} (\sqrt{(x + 9)} - 3) / x$.

(A) 0
(B) 1/6
(C) 1/3
(D) The limit does not exist

2. If $f(x) = x^2 \sin(3x)$, then $f'(x) =$

(A) $2x \cos(3x)$
(B) $2x \sin(3x) + x^2 \cos(3x)$
(C) $2x \sin(3x) + 3x^2 \cos(3x)$
(D) $6x \cos(3x)$

3. If $f(x) = (x^2 + 1)^3$, then $f'(1) =$

(A) 12
(B) 24
(C) 48
(D) 8

4. An equation of the line tangent to $y = x^3 - 2x$ at $x = 1$ is

(A) $y = x - 2$
(B) $y = x + 2$
(C) $y = 3x - 4$
(D) $y = -x$

5. $\int_0^2 (3x^2 - 4x + 1)\, dx =$

(A) 0
(B) 2
(C) 4
(D) 6

6. $\int x\, e^{x^2}\, dx =$

(A) $e^{x^2} + C$
(B) $x\, e^{x^2} + C$
(C) $2e^{x^2} + C$
(D) $\tfrac{1}{2}\, e^{x^2} + C$

7. If $F(x) = \int_1^{x^2} \sqrt{(t + 3)}\, dt$, then $F'(x) =$

(A) $2x \sqrt{(x^2 + 3)}$
(B) $\sqrt{(x^2 + 3)}$
(C) $2x \sqrt{(x + 3)}$
(D) $(2/3)(x^2 + 3)^{3/2}$

8. The average value of $f(x) = x^2$ on the interval $[0, 3]$ is
(A) 3
(B) 9/2
(C) 9
(D) 1

9. If $dy/dx = xy$ and $y(0) = 2$, then $y =$
(A) $2 + x^2/2$
(B) $2e^x$
(C) $2e^{x^2/2}$
(D) $e^{x^2/2} + 2$

10. A curve is defined by $x = t^2$ and $y = t^3$. At $t = 2$, $dy/dx =$
(A) 3/2
(B) 3
(C) 6
(D) 12

11. Which of the following series converges?
(A) $\sum_{n=1}^{\infty} 1/n$
(B) $\sum_{n=1}^{\infty} 1/\sqrt{n}$
(C) $\sum_{n=1}^{\infty} 1/n^2$
(D) $\sum_{n=1}^{\infty} n/(n + 1)$

12. The radius of convergence of $\sum_{n=1}^{\infty} (x - 2)^n / (n \cdot 3^n)$ is
(A) 1/3
(B) 2
(C) 3
(D) ∞

13. The Maclaurin series for e^{x^2} begins with
(A) $1 + x + x^2/2 + \dots$
(B) $1 + x^2 + x^3/3 + \dots$
(C) $x^2 + x^4 + x^6 + \dots$
(D) $1 + x^2 + x^4/2 + \dots$

14. $\int x \cos x \, dx =$
(A) $x \sin x - \cos x + C$
(B) $x \sin x + \cos x + C$
(C) $-x \sin x + \cos x + C$
(D) $\tfrac{1}{2} x^2 \sin x + C$

Part II: Free Response

Question 1. Limits and Continuity (9 points)

Let $f(x) = (x^2 - x - 6)/(x^2 - 9)$ for $x \neq 3$ and $x \neq -3$.

(a) Find $\lim_{x \to 3} f(x)$. Show the algebra. (2 points)

(b) Find $\lim_{x \to -3^-} f(x)$ and $\lim_{x \to -3^+} f(x)$. What does this tell you about the graph at $x = -3$? (3 points)

(c) How should $f(3)$ be defined so that f is continuous at $x = 3$? Justify using the definition of continuity. (2 points)

(d) A student claims the Intermediate Value Theorem guarantees a solution to $f(x) = 1$ on $[0, 2]$. Evaluate this claim. (2 points)

Question 2. Derivatives and Their Applications (9 points)

Let $f(x) = x^3 - 6x^2 + 9x + 1$.

(a) Find $f'(x)$. On what intervals is f increasing and on what intervals decreasing? Justify. (3 points)

(b) Find the x-coordinate and value of each relative extremum. Justify using f'. (2 points)

(c) Find the point of inflection and the intervals where f is concave up and concave down. (2 points)

(d) Find the absolute maximum and absolute minimum values of f on $[0, 4]$. Justify. (2 points)

Question 3. Accumulation and Integration (9 points)

Water flows into a tank at a rate $R(t) = 12 - 2t$ gallons per minute for $0 \le t \le 6$, where t is in minutes. Water leaves the tank at a constant 4 gallons per minute. At $t = 0$ the tank holds 30 gallons.

(a) Write and evaluate an integral for the total amount of water that flows into the tank from $t = 0$ to $t = 6$. Include units. (2 points)

(b) Is the amount of water in the tank increasing or decreasing at $t = 5$? Give a reason. (2 points)

(c) Write an expression for $V(t)$, the number of gallons in the tank at time t, and find $V(6)$. (2 points)

(d) At what time on $0 \le t \le 6$ is the amount of water in the tank greatest? Justify, and give that amount. (3 points)

Question 4. Power Series (9 points)

Let $f(x) = 1/(1 + x^2)$.

(a) Write the first four nonzero terms and the general term of the Maclaurin series for f. (2 points)

(b) Find the radius and interval of convergence. Show your work, including the endpoints. (3 points)

(c) Use your series for f to find the first three nonzero terms of the Maclaurin series for $g(x) = \arctan x$, where $g'(x) = f(x)$ and $g(0) = 0$. (2 points)

(d) Use the first three nonzero terms from part (c) to approximate $\arctan(1/2)$. Use the alternating series error bound to give a bound on the error. (2 points)